

Clustered Federated Learning with Adaptive Similarity for Non-IID Data

Technical and Critical Analysis

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Course: Data Mining & Machine Learning
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Reference: Yi *et al.* (2025), *Electronics*, 14(22), 4454

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Presentation Outline

1. Introduction: Problem Statement and Context
2. AS-CFL: Architecture and Innovations
3. Phase 1: Adaptive Initial Clustering
4. Phase 2: Dynamic Adjustment via G_k
5. Phase 3: Aggregation and Differential Privacy
6. Full AS-CFL Algorithm
7. Experimental Results
8. Critical Analysis and Perspectives
9. Conclusion

Federated Learning: Principle and Challenges

Quote from McMahan et al. (2017)

"Communication-efficient learning of deep networks from decentralized data"

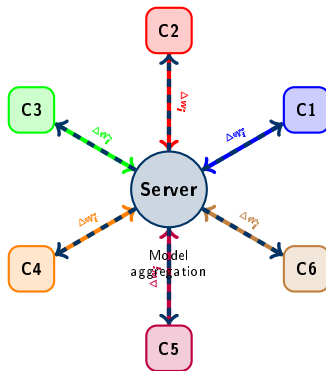
Principle of Federated Learning (FL):

- Distributed training without centralizing data
- Data stays local at each client
- Only model parameters are exchanged
- Compliant with GDPR and HIPAA

Critical applications:

- Healthcare: medical diagnosis
- Finance: fraud detection
- IoT: smartphones, sensors

Global model broadcast



The Major Challenge: Non-IID Data

Non-IID: *Non-Independent and Identically Distributed*

Label Distribution

Client A: 90% class 0

Client B: 90% class 1

→ Label skew

Feature Distribution

Day vs. night images

Different rotations

→ Feature skew

Quantity Imbalance

Client A: 10K samples

Client B: 100 samples

→ Quantity skew

Consequence: **Client Drift** (Zhao et al., 2018)

- Local updates diverge from the global objective
- Slower convergence, reduced accuracy, poor generalization
- Central problem of modern FL

State of the Art: Limitations of CFL Approaches

Method	Principle	Limitations
FedAvg	Simple global aggregation	Severe degradation with non-IID data
IFCA (Ghosh 2020)	Clustering with fixed K	Requires predefined K , unsuited to evolving data
FL+HC (Briggs 2020)	Hierarchical clustering	$O(N^2)$ complexity, non-scalable
CFL (Sattler 2019)	Cosine similarity	No quality control of clusters

Unmet Needs

- **Adaptability**: automatic adjustment of K
- **Scalability**: reduced computational complexity
- **Robustness**: quality-control mechanism

⇒ This is where **AS-CFL** comes in!

Adaptive Similarity Clustered Federated Learning

Objective

Partition S clients into K dynamic clusters to minimize:

$$F_k(w_k) = \sum_{i \in S_k} \frac{|D_i|}{|D_k|} F_i(w_k)$$

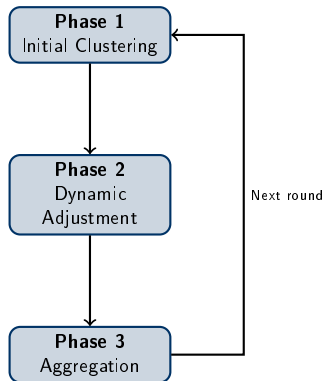
- Similarity computation
- Incentive

- Ratio G_k
- Split / Merge

4 Key Innovations

- 1 **Dynamic clustering** via G_k
- 2 **Positive incentive mechanism**
- 3 **Low-Rank Approximation**
- 4 **Differential privacy**

- DP noise
- Weighted aggregation



Phase 1A: Measuring Similarity Between Clients

Question: How can we tell if two clients have similar data without seeing their data?

Answer: Cosine Similarity on Updates

We compare the **direction** of the model's changes, not the data itself

$$\text{sim}(i, j) = \frac{\Delta w_i \cdot \Delta w_j}{\|\Delta w_i\|_2 \|\Delta w_j\|_2}$$

Intuitive interpretation:

- Δw_i = "direction" of client i
- If two clients move in the same direction $\rightarrow$ similar data
- Value between -1 and 1
(1 = identical, -1 = opposite)

Scalability problem:

- Full matrix: $S \times S$ clients
- Complexity: $O(S^2 d)$
- Infeasible with 1000+ clients!

Solution: Low-Rank SVD

- Approximation: $M \approx U \Sigma V^T$
- New complexity: $O(Srd)$

Phase 1B: Positive Incentive Mechanism (1/2)

Major Innovation of AS-CFL

Principle: Test Before Accepting

A client i joins cluster C_k **only if** its inclusion improves performance

Mathematical formulation:

$$\text{eval}(C_k, i) = \begin{cases} \text{True} & \text{if Accuracy}(w_{C_k \cup \{i\}}^k, D_{\text{test}}) > \theta \\ \text{False} & \text{otherwise} \end{cases}$$

Essential components:

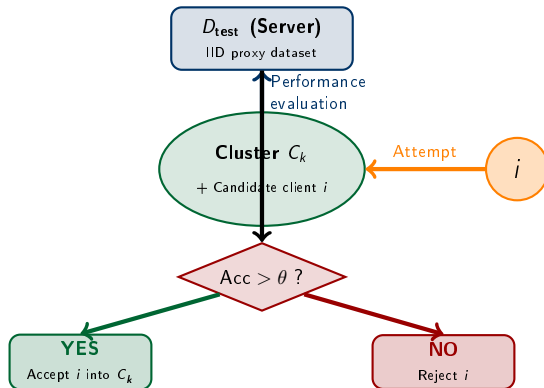
- D_{test} : IID proxy dataset
(small balanced set on the server)
- $\theta = 0.85$: acceptance threshold
(minimum required accuracy)
- $w_{C_k \cup \{i\}}^k$: temporary model
(test aggregation with the candidate)

Optimization strategy:

- 1 Select the top 20% most similar clients
- 2 Test each candidate sequentially
- 3 Stop if improvement $< 3\%$
- 4 Complexity: $O(S)$ forward passes

Phase 1B: Positive Incentive Mechanism (2/2)

Visual Decision Process



Key Benefit

Formal quality control: automatic filtering of outliers and divergent clients

⇒ Guarantees the consistency and quality of the clusters formed

Phase 1.C: Adjustment for Imbalance

Problem: Very heterogeneous dataset sizes across clients

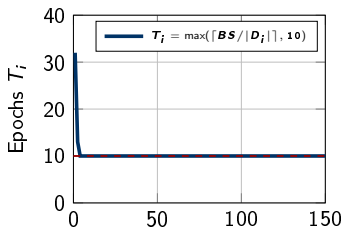
Solution: Adaptive Local Epochs

$$T_i = \max\left(\left\lceil \frac{BS}{|D_i|} \right\rceil, 10\right)$$

- T_i : number of epochs for client i
- $BS = 32$: fixed batch size
- Minimum of 10 epochs enforced

Numerical example:

- Client A: $|D_A| = 100 \Rightarrow T_A = 10$
- Client B: $|D_B| = 5000 \Rightarrow T_B = 10$
- Client C: $|D_C| = 20 \Rightarrow T_C = 10$



Phase 2A: EM Reassignment

Objective

Group similar movements together to identify **coherent update patterns**

EM Algorithm: Iterative Learning

- 1 **E-step (Expectation)**: assignment of movements
 - Each movement Δw_i is assigned to the closest cluster
 - Criterion: minimum distance to center w_k
- 2 **M-step (Maximization)**: recomputing centers
 - New center = average of the cluster's movements
 - $w_k = \frac{\text{sum of movements}}{\text{number of movements}}$

In practice

The algorithm converges quickly (5-10 iterations) toward stable groups

Phase 2B: Quality Evaluation: the Ratio G_k

Key question: How do we know if our clusters are of good quality?

The Ratio G_k : a Quality Indicator

$$G_k = \frac{d_{in}(k)}{d_{out}(k)} = \frac{\text{Internal dispersion}}{\text{External separation}}$$

Internal dispersion $d_{in}(k)$:

- Measures cluster homogeneity
- = average distance to the center
- Small = compact cluster

External separation $d_{out}(k)$:

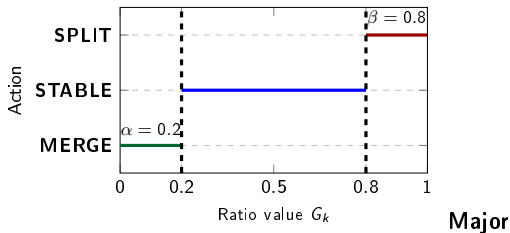
- Measures cluster isolation
- = distance to the nearest neighboring cluster
- Large = well separated

Interpretation

- G_k **low** → Good cluster (compact and isolated)
- G_k **high** → Problematic cluster (dispersed or too close to others)

Phase 2C: Dynamic Split/Merge Decisions

AS-CFL automatically adjusts the number of clusters based on G_k



Consequence: the number of clusters K evolves **automatically** every round $\rightarrow$ no predefinition needed!

Note: thresholds $\alpha = 0.2$ and $\beta = 0.8$ determined empirically (cross-validation)

Adjustment mechanism:

MERGE if $G_k < 0.2$

Why? Cluster too compact and isolated

- Find the nearest neighboring cluster
- Merge for greater robustness
- Recompute the new center

SPLIT if $G_k > 0.8$

Why? Cluster dispersed or too close to others

- Likely contains subgroups
- Apply K-Means ($K=2$) on members
- Create two new clusters

Phase 3: Aggregation and Differential Privacy

Final step: create the personalized model for each cluster

1. Weighted Aggregation:

Combine the cluster's updates:

$$w_k^{t+1} = w_k^t + \sum_{i \in S_k} \frac{|D_i|}{|D_k|} \cdot \Delta w_i'$$

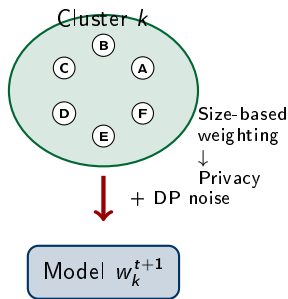
- Each client has a weight: $\frac{|D_i|}{|D_k|}$
- More data = more influence
- Result: a specialized model for the cluster

2. Privacy Protection (DP):

Adding Gaussian noise:

$$\Delta w_i' = \Delta w_i^t + \mathcal{N}(0, \sigma^2 I)$$

- $\mathcal{N}(0, \sigma^2)$: Gaussian noise
- Parameters: $\epsilon = 1.0, \delta = 10^{-5}$
- Guarantee: Differential Privacy



AS-CFL Algorithm: Full Structure

Inputs:

- Initial model w^0
- Set of clients S
- Number of rounds T

Main loop (each round):

1 Local training

- Each client trains its model
- Computes Δw_i (update)

2 Phase 1: Clustering

- Cosine similarity + Low-Rank
- Incentive mechanism

3 Phase 2: Adjustment

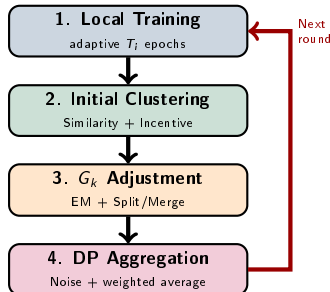
- EM algorithm
- Split/Merge via G_k

4 Phase 3: Aggregation

- Adding DP noise

Weighted aggregation

Cycle of a Round



Output: personalized models $\{w_k^T\}$, one per cluster

Experimental Setup

Datasets Used:

- **MNIST**: 10 classes
60K train, 10K test
- **EMNIST**: 62 classes
671K train, 112K test

Non-IID Simulation:

- **Dirichlet distribution**:
 $\alpha \in \{0.8, 0.6, 0.4\}$
(lower α = higher heterogeneity)
- **Imbalance**: 10%-60%
- **Feature bias**: rotations 0° , 90° , 180° , 270°

Hyperparameters:

- Learning rate: 0.1 (decay 0.99)
- Batch size: 32
- Local epochs: ~ 10 /round
- DP: $\epsilon = 1.0$, $\delta = 10^{-5}$

Baselines:

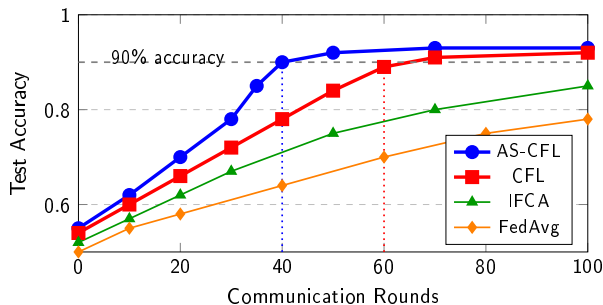
- **FedAvg**: single global model
- **CFL**: fixed clustering
- **IFCA**: predefined K

Environment:

- 20 clients
- 100 communication rounds
- PyTorch (GPU: NVIDIA RTX 3080)

Result 1: Convergence Speed

AS-CFL converges 20% faster than CFL



Key Observation (EMNIST, $\alpha = 0.6$)

AS-CFL reaches 90% in **40 rounds** vs. **60 rounds** for **CFL**

⇒ **33% reduction in communication!**

Result 2: Accuracy Under Heterogeneity

AS-CFL is more robust when data is highly heterogeneous

Dataset	Method	$\alpha=0.8$	$\alpha=0.6$	$\alpha=0.4$
MNIST	AS-CFL	0.917	0.915	0.910
	CFL	0.918	0.912	0.905
	IFCA	0.890	0.882	0.865
	FedAvg	0.875	0.850	0.805
EMNIST	AS-CFL	0.693	0.688	0.682
	CFL	0.695	0.685	0.675
	IFCA	0.620	0.605	0.580
	FedAvg	0.580	0.545	0.500

Key Observations:

- At **low heterogeneity** ($\alpha = 0.8$): AS-CFL $\approx$ CFL
- At **high heterogeneity** ($\alpha \leq 0.6$): **AS-CFL superior**
- **Graceful degradation**: AS-CFL maintains good performance even at $\alpha = 0.4$
- On EMNIST ($\alpha = 0.4$): AS-CFL (68.2%) vs. FedAvg (50.0%) = **+36% relative (statistically significant improvement, $p < 0.01$)**

Result 3: Ablation Study

Question: How important is each component?

Variant	Accuracy	Rounds (90%)	Δ Rounds
Full AS-CFL	0.915	40	—
Without Incentive	0.885	55	+37.5%
Without Dynamic Adjustment	0.902	48	+20%
Without Low-Rank	0.921	52	+30%

Analysis:

- **Incentive mechanism:** the most critical
3.3% accuracy loss and +37.5% rounds without it
- **Dynamic adjustment:** important for convergence
-1.4% accuracy due to suboptimal granularity
- **Low-rank:** accuracy/efficiency trade-off
+0.6% accuracy but +30% rounds → **approximation justified**

Strengths of the AS-CFL Approach

Methodological Innovations

- 1 **Truly dynamic clustering**: data-driven K via G_k
- 2 **Formalized incentive mechanism**: quality control unique in CFL
- 3 **Algorithmic scalability**: reduction from $O(S^2d)$ to $O(Srd)$
- 4 **Built-in privacy**: native differentially private noise

Solid Results:

- -20% communication rounds
- Robustness under strong non-IID conditions
- Graceful degradation
- Validated by ablation

Positioning:

- Combines adaptability + scalability
- Only method with quality control
- **New CFL standard**

Impact

AS-CFL balances **computational efficiency**, **communication efficiency**, and **personalization quality**

Identified Limitations

1. Limited Experimental Datasets

- Validation only on MNIST/EMNIST (relatively simple, 28×28 grayscale)
- Simulated non-IID patterns vs. natural heterogeneity
- Generalization to complex domains (medical, finance) still to be validated

2. Open Questions About the Incentive Mechanism

- Dependence on the proxy dataset D_{test} (hard to obtain in specialized domains)
- Computational cost (repeated forward passes)
- Empirical threshold $\theta = 0.85$, no automatic adaptation

3. Theoretical Gaps

- Thresholds $\alpha = 0.2$, $\beta = 0.8$ empirical, without theoretical justification
- **No formal convergence analysis**
- Impact of DP noise not precisely quantified

Future Research Perspectives

1. Methodological Extensions

- **Modern architectures:** adaptation to Transformers, fine-tuning with LoRA
- **Hierarchical clustering:** multi-scale (macro $\rightarrow$ micro $\rightarrow$ personal)
- **Advanced mechanisms:** Bayesian approach, RL (multi-armed bandit)

2. Theoretical Guarantees

- Convergence bounds under non-IID assumptions
- Formal proof of (ϵ, δ) -DP for the entire process
- Analysis of the cumulative privacy budget

3. Extended Validation

- Complex datasets: CIFAR-10/100, ImageNet, medical data (ISIC, ChestX-ray)
- Standard FL benchmarks: LEAF (FEMNIST, Shakespeare), FLSim
- Real-world applications: healthcare (diagnosis), finance (fraud), IoT (smart cities)

Summary and Conclusion

Key Contributions of AS-CFL

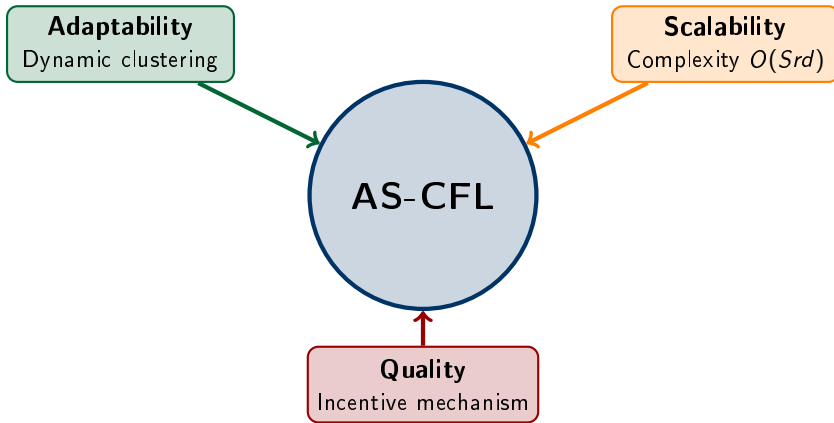
- 1 Dynamic clustering via ratio G_k : **automatic data-driven K**
- 2 Positive incentive mechanism: **formal quality control**
- 3 Low-rank approximation: $O(Srd)$ **scalability**
- 4 Differential privacy: **built-in protection**

Demonstrated Results

- **-20% rounds** of communication vs. CFL
- **Superior robustness** under strongly non-IID conditions
- **Graceful degradation** as heterogeneity increases

Impact and Vision

AS-CFL establishes a **new standard** for CFL, combining adaptability, scalability, and robustness. Future work will need to establish theoretical guarantees and validate on real-world applications for large-scale deployment.



AS-CFL: Toward Personalized, Efficient, and Safe FL

"A significant step toward global collaborative learning"

Thank you for your attention!

Questions and Discussion

References:

Yi et al. (2025), Electronics, 14(22), 4454

- ① Yi, G., Wu, Z., Zhang, X., Li, X. (2025). *Clustered Federated Learning with Adaptive Similarity for Non-IID Data*. Electronics, 14(22), 4454.
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